EXPERIMENT 2

CISCO PACKET TRACKER

AIM:

To study the packet tracer tool installation and user interface.

Description.

This experiment introduced cisco packet traces simulation tool and its user interface for designing and simulation computer network. It also covers the basics networking device such as hubs and switch. A hub broadcasts data to all connected device, whereas a switch forwards data only to the intended destination using mac addresses, making communication faster and more efficient.

Procedure:

1) install & launch cisco packet tracker

2) observe the user interfaces including the device toolbar & workspace.

3) Drag & place networking device such as PCs, hubs and switches onto the workspace.

4) Connect the device using the appropriate cables

5) assign Ip address to the PCs.

6) Test connectivity btw the device using ping command.

7) observe the data transmission through the hub and the switch

8) Save the network topology and record observation

OBSERVATION:

a) Behaviour of switch and hub in forwarding packet

Hub: A hub forwards the received data to all connected device without checking the destination address.

Switch: A switch forwards the received data only to the destination device by using MAC address.

b) Network topology implemented in the college.

Most colleges use a star topology where all computers are connected to central switch.

The hub broadcast packet to all connected device.

RESULT:

The cisco packet Tracker environment was successfully explored. The working of hubs and switch was studied and verified.